

llmXive Automated Review of Agentic Abstention: Do Agents Know When to Stop Instead of Act?

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript makes several specific quantitative claims regarding model performance and dataset composition that require verification against the provided text and tables.

First, the Introduction states the study evaluates “13 LLM-as-agent systems.” However, Section 4.1 lists 9 model families and 2 scaffolds. While the Qwen family has four sizes listed, the text does not explicitly clarify if these are counted as distinct “systems” to reach the total of 13. This ambiguity makes the claim difficult to verify without further clarification.

Second, Table 1 reports that “Llama-3.3-70B + 70B” achieves an AbsRec@10 of **100.0%** on WebShop. Given the dataset includes 101 held-out examples with challenging categories like “Missing Target” and “False Premise,” a perfect score is highly suspect. It suggests a potential calculation error, data leakage, or a mislabeled metric (e.g., perhaps it is 99.0% rounded). This claim is central to the paper’s contribution and requires immediate verification.

Third, the Results section claims Qwen3-235B achieves “0.59 AbsRec@1 and 0.71 AbsRec@10” in the QA scenario. These specific numbers are not found in the provided text body or tables (Table 1 focuses on WebShop; the appendix table focuses on AbstentionBench/TerminalBench but lists different values). The claim relies on a figure caption or an unprovided table, weakening the evidentiary support.

Finally, the citation for Terminal-Bench 2.0 (merrill2026terminal) lists the year as 2026. While this may be a future-dated preprint, it is an unusual citation for a benchmark used in a current study and should be double-checked for accuracy.

Action items.

- **[writing]** The claim of ‘13 LLM-as-agent systems’ (Intro) conflicts with the listed models in Sec 4.1. Clarify if parameter counts (e.g., Qwen 8B/235B) count as separate systems to reach 13, or correct the number.
- **[science]** Table 1 claims ‘Llama-3.3-70B + 70B’ achieves 100.0% AbsRec@10 on WebShop. This perfect score on 101 held-out examples is statistically improbable; verify for calculation errors, leakage, or rounding artifacts.
- **[writing]** The Results section cites Qwen3-235B scores (0.59/0.71) for QA, but these specific numbers are missing from the provided text and tables. Ensure they are explicitly reported to support the claim.
- **[writing]** The citation for ‘Terminal-Bench 2.0’ (merrill2026terminal) lists year 2026. Verify this is the correct version/year used, as future-dated citations for current benchmarks can be confusing.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively visualizes a specific ‘Missing Target’ abstention scenario with clear action timelines, but the provided caption is fragmented, contains raw formatting artifacts, and fails to accurately describe the specific content shown.

Figure 2

Figure 2 effectively illustrates the task components and instruction rewriting process. The diagram in (a) clearly labels the four core components, and the examples in (b) provide concrete instances of the two abstention scenarios described in the caption.

Figure 3

The figure effectively illustrates the claim that abstention recall increases with K, but the legend is poorly designed with duplicate markers/colors for different models, and the ‘Terminal’ subplot fails to display all models listed in the legend.

Figure 4

Figure 4 fails to match its caption’s description of a multi-row layout across scenarios; instead, it shows four unlabeled horizontal panels without scenario identification, and y-axis labels are missing from three of the four plots.

Figure 5

The figure effectively displays Abstention Recall across four categories, but the caption’s claim that ‘Missing Target’ is the hardest case for all models is contradicted by the data for the top-performing model (Llama-3.3-70B), which achieves significantly higher recall in that category than in others.

Figure 6

The figure displays three panels of Abstention Recall vs K, but the x-axis label is missing for the first two panels, and the legend placement is ambiguous.

Figure 7

Figure 7 is critically flawed because it lacks a caption, preventing verification of the data and metrics. Additionally, the metric ‘Over-Abstention Rate’ is not defined in the context of the paper’s main ‘Abstention Recall’ focus, and the legend placement is suboptimal.

Figure 8

The figure is fundamentally broken as the rendered image (four specific abstention categories) contradicts the caption’s description (three scenarios: Web, Terminal, QA). The visual content does not support the claims made in the caption regarding the layout or the specific scenarios analyzed.

Figure 9

The figure is severely mismatched with its caption; the image shows four specific abstention categories while the caption describes a comparison across three different scenarios (Web, Terminal, QA) and mentions a category (‘Missing Prerequisite’) not present in the plot.

Figure 10

The figure is incomplete, showing only panel (a) while the caption describes panels (b) and (c) and contains a broken cross-reference. Additionally, the overlapping bars in the histogram obscure direct comparison between the two distributions.

Figure 11

The figure is incomplete, displaying only panel (a) despite the caption describing three panels (a, b, c). Additionally, the visual data in panel (a) contradicts the caption’s claim of similar token counts, and the caption contains a broken cross-reference.

Figure 12

The figure is a t-SNE plot with a specific title for ‘Subjective Preference’, but it lacks a legend to explain the color coding of the points. Additionally, the caption implies the figure covers three categories, while the image only shows one labeled panel.

Action items.

- **[writing]** Figure 1: The provided caption text is fragmented and does not match the visual content; it discusses ‘web shopping, terminal interactions, and question answering’ generally, whereas the figure specifically illustrates a single ‘Missing Target Task’ scenario with a timeline of agent actions.
- **[writing]** Figure 1: The caption contains raw LaTeX formatting artifacts (e.g., ‘%’, ‘[overview.pdf]’) and disjointed sentence fragments that should be cleaned up for the final version.
- **[writing]** Figure 3: The legend at the top lists 10 models, but the ‘Terminal’ subplot only displays 4 distinct lines, making it impossible to identify which models are missing or which lines correspond to the unlisted models.
- **[writing]** Figure 3: The legend uses identical green diamond markers and similar green line colors for both ‘gpt-oss-120b’ and ‘Terminus 2’, creating ambiguity in distinguishing these

two series.

- **[fatal]** Figure 4: The caption describes a multi-row layout ('From top to bottom, the rows show Web, Terminal, and QA results'), but the rendered image displays four horizontally arranged panels labeled 'Missing Target', 'Subjective Preference', 'Underspecified Intent', and 'False Premises' with no row structure or scenario labels.
- **[science]** Figure 4: The caption claims to show results across Web, Terminal, and QA scenarios, but the figure lacks any visual indicators (e.g., panel titles, axis labels, or legends) identifying which data corresponds to which scenario.
- **[writing]** Figure 4: The x-axis label 'K' is present, but the y-axis label 'Abstention Recall' is only visible on the leftmost panel; the other three panels lack y-axis labels, reducing clarity.
- **[science]** Figure 5: The caption claims 'Missing Target... is the hardest case for all models,' but the 'Missing Target' panel shows Llama-3.3-70B achieving ~ 0.62 AbsRec@1, whereas 'False Premises' shows the same model at ~ 0.95 . The 'hardest' category (lowest recall) is actually 'Missing Target' only for the lower-performing models, while the top model finds it relatively easy compared to other categories; the caption's absolute claim contradicts the visual data for the best-performing model.
- **[writing]** Figure 5: The y-axis label 'Abstention Recall' is present, but the caption uses the notation '*AbsRec@K*' without explicitly defining that the y-axis represents this specific metric, though it is implied.
- **[science]** Figure 6: The x-axis label 'K' is positioned only under the rightmost subplot, leaving the x-axes of the left and middle subplots unlabeled.
- **[writing]** Figure 6: The legend is placed outside the plot area at the bottom center, which is ambiguous regarding which subplot it applies to; it should be placed inside or clearly associated with the specific panel.
- **[fatal]** Figure 7: The figure has no caption provided (labeled '(no caption)'), making it impossible to verify what the plotted data represents or to interpret the specific models and metrics shown.
- **[science]** Figure 7: The legend lists 'GPT-5.4-mini' variants, but the y-axis is 'Over-Abstention Rate' while the paper's other figures (e.g., Fig 3, 4) focus on 'Abstention Recall'; the metric definition and its relationship to the paper's core claims are unclear without a caption.
- **[writing]** Figure 7: The legend is placed outside the plot area without a border or background, which can be visually confusing and makes it harder to associate the symbols with the lines compared to an inset legend.
- **[fatal]** Figure 8: The rendered image displays four panels labeled 'Missing Target', 'Subjective Preference', 'Underspecified Intent', and 'False Premises', but the caption describes a triptych of 'Web, Terminal, and QA scenarios' with rows. The visual content does not match the caption's description of the layout or the specific scenarios presented.

- **[science]** Figure 8: The caption claims to show results for ‘Missing Prerequisite in Terminal’, but the rendered panels do not contain a ‘Missing Prerequisite’ category or a ‘Terminal’ scenario label, creating a disconnect between the text and the data shown.
- **[fatal]** Figure 9: The rendered image displays four panels labeled ‘Missing Target’, ‘Subjective Preference’, ‘Underspecified Intent’, and ‘False Premises’, but the caption describes a triptych of ‘Web, Terminal, and QA scenarios’ and mentions ‘Missing Prerequisite’ in Terminal, which is not shown. The figure content does not match the caption.
- **[science]** Figure 9: The y-axis is labeled ‘Abstention Recall’ but the caption refers to ‘*AbsRec@K*’. While likely the same metric, the axis label should explicitly match the caption’s notation or define the relationship to avoid ambiguity.
- **[writing]** Figure 10: The caption references panels (b) and (c) and a missing Figure number (‘shown in Figure .’), but the provided image contains only panel (a).
- **[science]** Figure 10: The histogram bars are overlapping rather than side-by-side or transparent, making it difficult to compare the ‘Original’ and ‘Rewritten’ distributions at specific token lengths.
- **[fatal]** Figure 11: The caption describes three panels (a, b, c), but the rendered image only shows panel (a) ‘Token Length Distribution’; panels (b) and (c) are missing.
- **[science]** Figure 11: The caption claims ‘similar token counts’ between original and rewritten instructions, but the histogram shows a clear shift where ‘Rewritten’ (orange) has a longer tail and higher values than ‘Original’ (blue).
- **[writing]** Figure 11: The caption contains an incomplete cross-reference: ‘shown in Figure .’ lacks the specific figure number.
- **[science]** Figure 12: The plot displays two distinct colors (blue and orange) for data points, but the caption and plot lack a legend defining what these colors represent (e.g., original vs. rewritten instructions).
- **[writing]** Figure 12: The title ‘(a) Subjective Preference’ suggests this is a sub-panel, yet the caption describes the figure as containing visualizations for three categories (‘Subjective Preference, Underspecified Intent, and False Premise or Contradiction’), creating a mismatch between the single panel shown and the multi-category description.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on field-specific jargon and undefined acronyms that create

barriers for non-specialist readers. The most critical issue is the use of the macro `\method` in the Introduction (Section 1) and throughout the text without a prior definition. The reader is forced to wait until Section 5 to learn that `\method` stands for “Context Evolution.” This should be defined immediately upon first introduction, e.g., “We introduce Context Evolution (`\method`), a method that...”

In Section 2, the paper formulates the problem as a “POMDP” without spelling out “Partially Observable Markov Decision Process.” While standard in reinforcement learning, this excludes readers from adjacent fields. Similarly, “SPL” is introduced in Section 4.2 as a metric without expansion; it should be written as “Success weighted by Path Length (SPL)” at first use.

The term “scaffolds” (Section 4.1) is used to describe agent frameworks (e.g., Codex CLI, Terminus 2). This is becoming common jargon in the agentic AI subfield but is opaque to a broader audience; “frameworks” or “orchestration layers” would be clearer. Additionally, the caption for Figure 1 in the appendices (e001) references “t-SNE” without defining the acronym.

Finally, the distinction between “Request-based” and “Environment-based” abstention is introduced with heavy reliance on these specific labels. While necessary for the taxonomy, the text could benefit from a brief plain-English gloss (e.g., “infeasible from the start” vs. “infeasible revealed during interaction”) to ensure clarity for readers unfamiliar with the specific dataset construction logic.

Action items.

- **[writing]** Define the macro `\method` at its first occurrence in the Introduction. Currently, it appears as a command without a preceding definition, forcing readers to guess it stands for ‘Context Evolution’ until Section 5.
- **[writing]** Replace the acronym ‘POMDP’ in Section 2 with ‘Partially Observable Markov Decision Process (POMDP)’ or a plain English description for non-specialist readers.
- **[writing]** Define ‘SPL’ (Success weighted by Path Length) at first use in Section 4.2. While standard in robotics, it is jargon to general NLP readers and requires a brief plain-language explanation.
- **[writing]** Replace the term ‘scaffolds’ in Section 4.1 with ‘frameworks’ or ‘orchestration layers’ to reduce field-specific jargon.
- **[writing]** Define ‘t-SNE’ in the caption of Figure 1 (e001) or the main text. While common, it is an acronym that should be spelled out for a general audience.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The manuscript presents a coherent argument for the necessity of “Agentic Abstention” as a

distinct capability from single-turn LLM abstention. The logical flow from problem definition to benchmark construction and evaluation is generally sound. However, there are specific inconsistencies in the definition and reporting of metrics that undermine the internal consistency of the results.

First, the definition of “Timely Recall” is contradictory. In Section 4.2 (Evaluation Metrics), the authors explicitly define Timely Recall as AbsRec@1 for request-based tasks and AbsRec@2 for environment-based tasks, acknowledging that the “warranted step” (w_i) differs (immediate vs. after one interaction). However, in Section 5.1 (Results) and Table 1, the authors report a single “Timely Recall” value for the WebShop benchmark (26.7% for Llama-3.3-70B). Since WebShop contains 251 environment-based tasks (Missing Target), the reported metric should logically be AbsRec@2 if the definition in Section 4.2 is strictly followed. Reporting AbsRec@1 for these tasks would penalize agents for not abstaining *before* the environment reveals the impossibility, which contradicts the definition of w_i for environment-based tasks. This inconsistency makes it impossible to verify if the reported “Timely Recall” figures are comparable across the different task categories.

Second, the causal claim regarding model scale is not fully supported by the presented evidence. The Introduction states that “larger or more capable models sometimes perform worse at timely abstention.” While the paper notes that scaling improves overall recall, the evidence for the negative impact on *timely* recall is mixed and confounded by the benchmark. For instance, in the QA scenario, the largest model (Qwen3-235B) achieves the highest timely recall (0.59), whereas in WebShop, the smaller Llama-8B has a lower timely recall than Llama-70B. The claim that larger models perform *worse* is not consistently demonstrated across the reported data points; rather, the data suggests that performance is highly dependent on the specific task category (e.g., Missing Target vs. False Premise) rather than a simple inverse relationship with model scale. A more nuanced discussion or a controlled comparison is needed to support this specific causal assertion.

Finally, there is a slight logical tension between the treatment of “clarification” as “abstention” and the dataset construction. The authors define clarification requests as instances of ABSTAIN (Section 2). However, in the Interactive QA setup (Section 3), they explicitly exclude datasets like MuSiQue and SQuAD 2.0, which often require clarification or multi-hop reasoning where the answer is not immediately obvious. By removing these cases, the evaluation set may bias the results towards scenarios where abstention is the *only* valid stop signal, potentially inflating the perceived difficulty of “timely” abstention or masking the agent’s ability to distinguish between “I need more info” (clarify) and “I cannot solve this” (abstain). This limits the generalizability of the conclusion that agents struggle to know *when* to stop, as the “stop” signal in the remaining data is binary and absolute.

Action items.

- **[writing]** The manuscript presents a coherent argument for the necessity of “Agentic Abstention” as a distinct capability from single-turn LLM abstention. The logical flow from problem definition to benchmark construction and evaluation is generally sound. However, there are specific inconsistencies in the definition and reporting of metrics that undermine the internal consistency of the results. First, the definition of “Timely Recall” is contradictory.

In Section 4.2 (Evaluation Metrics), the authors exp

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extend slightly beyond the immediate evidence provided, particularly regarding the generalizability of scaling effects and the robustness of the proposed method’s improvements.

First, the Introduction states that “larger or more capable models sometimes perform worse at timely abstention.” While the results section (Section 6.2) does show that Qwen3-235B has lower timely recall than smaller Qwen variants in specific instances, the paper does not provide a statistical analysis to support this as a general trend across the evaluated models. The data is sparse (only a few models per family), and the effect might be specific to the Qwen architecture or training data rather than a universal property of model scale. The claim should be qualified to reflect that this is an observed phenomenon in specific cases rather than a broad scaling law.

Second, the proposed method, \method, is claimed to significantly improve timely recall (e.g., from 26.7% to 57.4% for Llama-3.3-70B on WebShop) based on training on only 20 trajectories. While the improvement is notable, the small training set size raises concerns about overfitting and the stability of the result. The paper does not report variance, confidence intervals, or results from multiple random seeds, which are necessary to substantiate the claim that the method is robust and data-efficient. The conclusion should be tempered to acknowledge the limited training data and the need for further validation on larger datasets.

Third, the claim that “lessons from smaller models (8B) transfer effectively to larger models (70B)” is supported by the data in Table 2, but the magnitude of the transfer (55.3% vs. 57.4% for 70B-to-70B) is relatively small. The phrasing “transfer effectively” suggests a strong generalization capability that the data only partially supports. A more nuanced discussion of the transferability, perhaps highlighting the specific conditions under which it works best, would be more accurate.

Finally, the definition of “Timely Recall” is inconsistent across benchmarks: AbsRec@1 for request-based tasks and AbsRec@2 for environment-based tasks. This inconsistency makes direct comparisons between benchmarks difficult and risks over-claiming the comparability of results. The paper should either unify the metric definition or provide a clear justification for why a delay of one step is acceptable in environment-based tasks but not in request-based tasks.

Action items.

- **[writing]** The paper makes several claims that extend slightly beyond the immediate evidence provided, particularly regarding the generalizability of scaling effects and the robustness of the proposed method’s improvements. First, the Introduction states that “larger or more capable models sometimes perform worse at timely abstention.” While the results section

(Section 6.2) does show that Qwen3-235B has lower timely recall than smaller Qwen variants in specific instances, the paper does not provide a stat

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper addresses a critical safety-adjacent capability: Agentic Abstention, defined as the ability of an agent to recognize when a task is infeasible and stop acting. This is a positive contribution to safety, as it directly targets the reduction of hallucinated completions and unnecessary tool use, which can lead to resource exhaustion or unintended side effects in real-world deployments.

However, from a safety and ethics lens, the manuscript requires minor revisions to fully address the implications of its findings and methodology:

1. **High-Stakes Deployment Risks:** The “Broader Impacts” section (Section 7) correctly identifies that over-abstention can frustrate users. However, it does not sufficiently address the risks in high-stakes domains (e.g., medical diagnosis, legal advice, or autonomous driving). In these contexts, an agent that fails to abstain (false negative) could cause direct harm, while an agent that abstains too readily (false positive) could deny necessary services. The authors should explicitly state that the current benchmarks and the proposed \method are not sufficient for deployment in such high-risk environments without human-in-the-loop oversight. A sentence clarifying the “human oversight” requirement for critical applications is necessary.
2. **Adversarial Dataset Construction:** The construction of the WebShop and Terminal-Bench abstention datasets relies heavily on LLMs (specifically GPT-5.4-mini) to rewrite instructions into “False Premise” or “Underspecified” categories (Section 3, Appendix D). While the authors mention a review process, there is no explicit confirmation that these generated adversarial prompts were screened for safety. There is a risk that the rewriting process could inadvertently introduce toxic, biased, or sensitive content into the test set. The authors should add a statement confirming that a human review or safety filter was applied to the generated adversarial examples to ensure they do not contain harmful content.
3. **Sensitive Benchmark Integration:** The evaluation includes datasets like MediQ (medical questions) and BBQ (bias benchmark) (Section 3, Appendix D). The paper focuses on “unanswerability” but does not explicitly discuss how the agents handled sensitive or potentially harmful queries within these datasets. Did the agents abstain on sensitive topics as a safety measure, or did they attempt to answer? The authors should clarify if the evaluation protocol included specific safety constraints for these subsets or if the “ABSTAIN” action was the only safe response for sensitive topics. This distinction is crucial

for understanding the agent’s safety alignment.

Overall, the paper presents a valuable framework for improving agent reliability, but the safety implications of its methodology and the limitations of its applicability to high-risk scenarios need to be more explicitly articulated.

Action items.

- **[writing]** The ‘Broader Impacts’ section (Section 7) acknowledges over-abstention risks but lacks a concrete mitigation strategy for high-stakes domains (e.g., medical or legal agents). Explicitly state that the proposed abstention mechanism is not a substitute for human oversight in critical applications and define specific failure modes where abstention could cause harm (e.g., denial of service).
- **[writing]** The dataset construction for WebShop and Terminal-Bench involves rewriting instructions to create ‘False Premise’ or ‘Underspecified’ scenarios (Section 3, Appendix D). While reviewed by authors, the process relies on LLMs (GPT-5.4-mini) to generate adversarial examples. Add a statement confirming that human verification was performed on a random sample of these generated adversarial prompts to ensure they do not inadvertently contain harmful, biased, or sensitive content.
- **[writing]** The evaluation includes medical datasets (MediQ) and bias benchmarks (BBQ) within the Interactive QA tasks (Section 3, Appendix D). The paper does not explicitly mention whether the evaluation protocol included safety filters or if the ‘ABSTAIN’ action was the only allowed response for sensitive topics. Clarify if the agents were evaluated on their ability to refuse harmful queries in these specific subsets, or if the focus was solely on factual unanswerability.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a comprehensive empirical study of “Agentic Abstention” across three distinct environments (Web, Terminal, QA) with a total sample size exceeding 28,000 instances. The scale of the evaluation, particularly the 27,073 samples from AbstentionBench, provides a strong foundation for the general claims regarding the difficulty of abstention. The definition of metrics (AbsRec@K) is mathematically rigorous and well-suited to the sequential nature of the problem.

However, the scientific evidence supporting the specific comparative claims regarding model scaling and the efficacy of the proposed method is weakened by a lack of statistical rigor. In the “Results” section, the authors claim that “larger or more capable models sometimes perform worse at timely abstention.” While the point estimates in the text and tables (e.g., Qwen3-235B vs. smaller variants) suggest this trend, the paper does not report confidence intervals, standard

deviations, or p-values for these comparisons. Given the stochastic nature of LLM inference and the relatively small sample size of the WebShop abstention subset ($n=500$), it is impossible to determine if these differences are statistically significant or merely noise.

Furthermore, the evaluation of the proposed method (Context Evolution) relies on a very small training set (20 trajectories) and a held-out test set of 101 examples. The dramatic improvement reported (e.g., AbsRec@1 rising from 26.7% to 57.4% for Llama-3.3-70B) is compelling but risks overfitting to the specific 20 examples used for playbook generation. The authors should demonstrate that these gains are robust by reporting results across multiple random seeds for the 20-example selection or by performing a cross-validation analysis. Without this, the claim that the method “distills reusable stopping rules” is not fully supported by the evidence provided.

Finally, the ground truth for the “Request-based Abstention” tasks in WebShop and Terminal-Bench is generated via an LLM (GPT-5.4-mini) rewriting process. While the authors provide t-SNE visualizations to show semantic overlap, they do not provide a human evaluation of the “abstain-warranted” labels. If the rewriting model fails to create truly unresolvable tasks or introduces subtle cues that make the task solvable, the evaluation metrics (AbsRec) would be measuring the agent’s ability to detect LLM-generated artifacts rather than genuine infeasibility. A human audit of a subset of these rewritten instructions is necessary to validate the dataset quality.

Action items.

- **[science]** The claim that ‘larger or more capable models sometimes perform worse at timely abstention’ (Intro) lacks statistical validation. The paper reports point estimates (e.g., Qwen3-235B vs smaller variants) but provides no confidence intervals, standard errors, or significance tests (e.g., t-tests) to confirm these differences are not due to variance, especially given the small sample size of the WebShop abstention set ($n=500$).
- **[science]** The proposed method (Context Evolution) is evaluated on a very small held-out set (101 examples) with only 20 training trajectories. The reported jump in AbsRec@1 from 26.7% to 57.4% (Table 1) risks overfitting to this specific distribution. The authors should report performance variance across multiple random seeds or provide a cross-validation analysis to demonstrate robustness.
- **[science]** The dataset construction for ‘Request-based Abstention’ relies on GPT-5.4-mini to rewrite instructions (Section Datasets). The paper claims these are ‘semantically indistinguishable’ (Figure 1) but does not provide a human evaluation of the ‘abstain-warranted’ labels. If the rewriting model introduces subtle biases or if the ‘unsolvable’ nature is ambiguous, the ground truth labels may be noisy, invalidating the AbsRec metrics.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in this paper is comprehensive in scope but lacks necessary rigor in reporting uncertainty and significance testing. While the definitions of metrics like AbsRec@K and SPL are mathematically sound (Appendix C), the presentation of results relies almost exclusively on point estimates without measures of variance.

Specifically, in Table 1 (WebShop results) and Table 2 (AbstentionBench/TerminalBench), the authors report precise percentages (e.g., 57.4%, 68.9%) derived from a held-out set of 101 examples. For a sample size of $N=101$, the standard error for a proportion near 0.5 is approximately 5%. Without reporting confidence intervals (e.g., 95% CI) or standard deviations, it is impossible to determine if the observed differences between models (e.g., the jump from 26.7% to 57.4% with \method) are statistically significant or if the “best” model is merely a result of random variance in the test set.

Furthermore, the paper makes strong comparative claims (e.g., “Codex CLI consistently achieves higher abstention recall than Terminus 2”) without citing the results of statistical hypothesis tests. Given the binary nature of the outcome (abstain vs. not abstain) at the episode level, appropriate tests such as McNemar’s test for paired comparisons or a generalized linear mixed model (GLMM) with random effects for the task instance should be employed. The absence of p-values or effect sizes with confidence bounds weakens the empirical support for the conclusion that specific scaffolds or model scales are definitively superior.

Finally, the study involves a large number of comparisons across 13 models, multiple environments, and various abstention categories. The manuscript does not address the multiple-comparisons problem. Without a correction method (e.g., False Discovery Rate control), the risk of false positives in identifying “hardest” categories or “best” models is non-negligible. The authors should either apply a correction and report adjusted p-values or explicitly frame their findings as exploratory rather than confirmatory.

Action items.

- **[science]** Report confidence intervals or standard errors for all primary metrics (AbsRec@K, SPL) in Tables 1 and 2. The current point estimates (e.g., 57.4% timely recall) lack measures of variance, making it impossible to assess the statistical significance of the reported improvements or the stability of the results across the 101 held-out examples.
- **[science]** Clarify the statistical test used to compare models (e.g., Llama-3.3-70B vs. Qwen3-235B) and scaffolds. The paper claims significant differences (e.g., scaffold effects) but does not specify if paired tests (e.g., McNemar’s or Wilcoxon signed-rank) were applied to the episode-level outcomes, nor does it report p-values.
- **[science]** Address the multiple-comparisons problem. With 13 models, 2 scaffolds, 3 environments, and 5+ abstention categories, the number of pairwise comparisons is large. The paper does not mention any correction method (e.g., Bonferroni, Holm-Bonferroni, or FDR) to control for Type I errors when interpreting the ‘best’ performing models.
- **[writing]** Provide the exact sample size (N) for each specific sub-category analysis (e.g.,

‘Missing Target’ in WebShop). While total counts are given (e.g., 251 tasks), the breakdown of solvable vs. unsolvable per specific sub-category in the results section is sometimes ambiguous, which affects the denominator for recall calculations.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a generally high standard of academic writing, with a clear logical flow from the problem definition to the proposed solution and results. The introduction effectively sets the stage for “Agentic Abstention,” and the distinction between single-turn and sequential abstention is articulated clearly. However, several areas require minor polishing to enhance precision and readability.

First, in the Introduction, the description of the proposed method (\method) contains a slight syntactic ambiguity: “distills full interaction trajectories into reusable stopping rules.” While the meaning is recoverable, the phrasing could be misinterpreted as the trajectories themselves being distilled. A more precise phrasing would be “distills reusable stopping rules *from* full interaction trajectories.”

Second, in Section 3 (Datasets), the sentence describing the WebShop adaptation is a long compound sentence that could be split for clarity: “We adapt WebShop... using the first 500 instructions as solvable instances and constructing 500 abstention-warranted instances.” Breaking this into two sentences or using a semicolon would improve the rhythm and reduce cognitive load for the reader.

Third, in Section 5.2, the claim “More reasoning improves timely recall but can reduce overall recall” is presented without an immediate subject for the second clause. While the context implies this refers to specific models (like Qwen3-235B), explicitly naming the model or setting in the sentence would prevent ambiguity.

Finally, the Appendix contains several passive constructions (e.g., “All generated instructions are reviewed”) that could be strengthened with active voice (“We reviewed all generated instructions”) to make the methodology description more direct and engaging. These are minor issues that do not obscure the scientific contribution but, if addressed, would elevate the overall polish of the paper.

Action items.

- **[writing]** In the Introduction, the phrase ‘distills full interaction trajectories into reusable stopping rules’ is slightly ambiguous. Clarify if the method distills rules *from* the trajectories or distills the trajectories *into* rules. Suggest: ‘distills reusable stopping rules from full interaction trajectories’.
- **[writing]** In Section 3 (Datasets), the sentence ‘We adapt WebShop... using the first 500

instructions as solvable instances and constructing 500 abstention-warranted instances’ is a run-on. Split into two sentences or use a semicolon to separate the two distinct actions for better readability.

- **[writing]** In Section 5.2 (Factors Impacting Performance), the sentence ‘More reasoning improves timely recall but can reduce overall recall’ lacks a clear subject for the second clause. Specify which model or setting exhibits this behavior (e.g., ‘...but can reduce overall recall for *Qwen3-235B*’).
- **[writing]** In the Appendix (Section ‘Adapting and Constructing...’), the phrase ‘All generated instructions are reviewed to ensure realistic abstention scenarios’ is passive. Consider active voice: ‘We reviewed all generated instructions to ensure...’ for stronger flow.